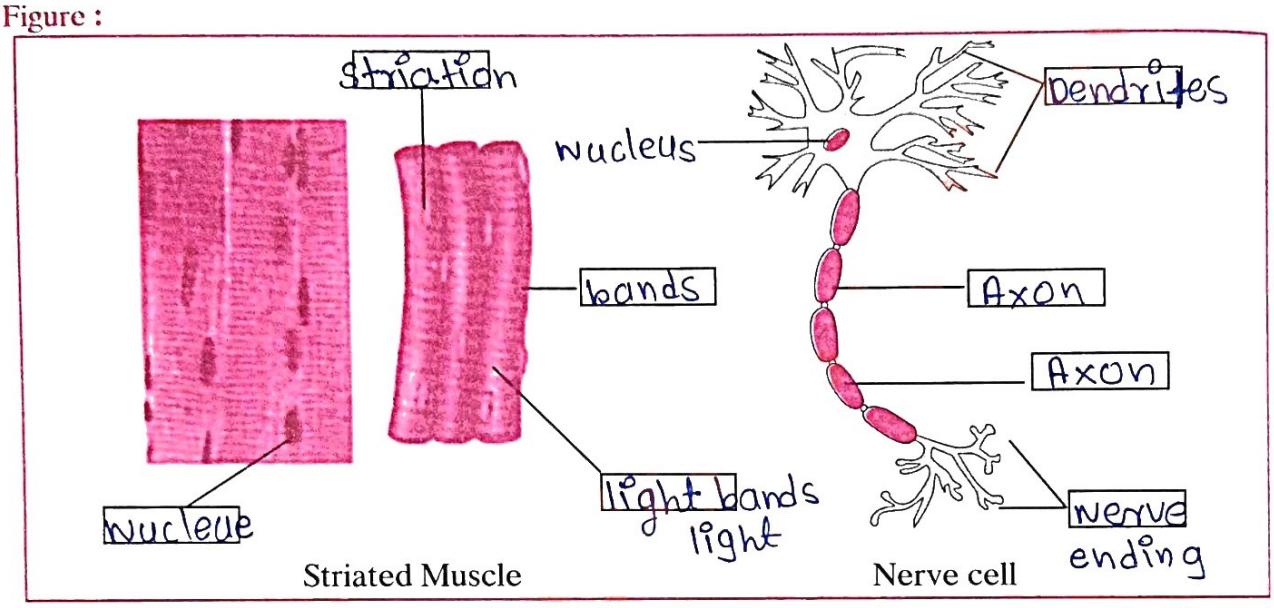
Experiment No.12

Title : **To identify striated muscle fibre and nerve cells in animal tissues from the prepared slides and draw their well labelled diagrams.**

Apparatus : Permenent slides and charts of striated muscle fibre and nerve cells in animal tissue , Microscope, etc.

Figure :



Procedure :

1. In animal tissues, take permenent prepared slides of striated muscle fibre and nerve cells.
2. Observe them under high magnification of compound microscope.
3. Record the observations in the given table.
4. Draw neat and labelled diagrams of striated muscle and nerve cells.

Observation Table : Striated Muscle fibre

No	Features	Observations
1	Size and shape	long, cylindrical, multinucleate.
2	Interconnections between fibres	multinucleate.
3	Branched / Unbranched	unbranched.
4	Dark and light bands	Present Alternately.
5	Working capacity (Fatigueness)	move as per our will
6	Nucleus	Present at the centre.
7	Voluntary/ Involuntary	voluntary muscles.

Occurance : Muscles of hand and legs, Abdominal, Tongue, Food pipe, Pharynx.

Functions : Being strong and powerful, provide force for locomotion and all other voluntary movements so they are also called skeletal muscles.

Neuron/ Nerve cells

No	Features	Observations
1	Shape and size	like branched tree
2	Nucleus	spherical, present at centre
3	Structure Main parts of cell	Dendrites :- Numerous, small. Axon :- Extremely long. Nucleus :- It is the main part of the nerve cell.

Functions : Transmission of impulses, Dendrites, receive impulses and the Axion takes the impulses away from the cell body.

Multiple Choice Questions

1. Striated Muscles are.....
   A. uninucleated ✓B. multinucleated C. binucleated D. anucleated
2. Presence of pits is a charactristic feature of .....
   ✓A. Parenchyma B. Collenchyma C. Sclerenchyma D. Aerenchyma
3. Myclin sheath is present in .....
   A. nerve cell B. Striated muscle fibre C. parenchyma D. cardiac muscles
4. Parenchyma can be identified by .....
   A. position of nucleus B. size of the cells
   ✓C. thickness of the cell wall D. absence of intracellular gaps
5. The striated muscles are .....shaped.
   A. Brick ✓B. Rod C. Disc D.Cuboidal

Questions

1. In reflex action, which parts of the nervous system is involved?
   In reflex action, the spinal cord along with the brain are involved.

2. Explain the importance of involuntary muscles in human body.
   1) one very important involuntary muscle is heart which keep beating all day and night.
   2) other involuntary muscles help digest food and are found in your stomach and the intestine. It takes gut to be an involuntary muscle.

H7YNH6